

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION

CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27

SET B — ADVANCED HOTS & ANALYTICAL THINKING**CHAPTER 03: ATMOSPHERE AND CLIMATE**

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-B-03

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET B - Advanced HOTS & Analytical Thinking):

1. The question paper comprises **25 Higher Order Thinking Skills (HOTS) Questions** of **1 mark each**.
2. Designed to test **deep causal reasoning, critical source evaluation, constitutional mechanics, and economic analysis**.
3. All questions are compulsory. There is no negative marking for incorrect answers.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

 **ADVANCED HOTS INSTRUCTIONS:** Carefully analyze the socio-historical contexts, geomorphic cause-and-effect relationships, and constitutional principles before selecting your answer.

Q.1 Why is the environmental lapse rate in the troposphere (~6.5 °C per km) positive (temperature falls with height), whereas in the stratosphere the temperature rises with increasing altitude?

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| (A) The troposphere is heated from below by terrestrial radiation, whereas the stratosphere contains ozone that absorbs incoming solar ultraviolet radiation directly | (B) The troposphere has no air molecules to hold heat |
| (C) The stratosphere is located closer to the center of the earth | (D) The sun only shines on the stratosphere and never on the troposphere |

Q.2 How does the Coriolis force alter the path of winds moving from the Subtropical High Pressure Belts towards the Equatorial Low Pressure Belt in the Northern Hemisphere?

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| (A) It deflects winds to their right, transforming a pure southward wind into the Northeast Trade Winds | (B) It deflects winds to their left, creating Southeast Trade Winds |
| (C) It forces winds to blow straight northwards into the Arctic | (D) It eliminates all atmospheric wind movement completely |

Q.3 Why is the Indian summer monsoon characterized by a dramatic seasonal 'burst' or sudden onset rather than a gradual change in weather?

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| (A) The sudden northward shift of the subtropical westerly jet stream beyond the Himalayas allows the tropical easterly jet stream to establish over peninsular India, abruptly pulling the moist southwest monsoon trough | (B) The Indian ocean evaporates completely within 24 hours |
| (C) The Himalayas flatten out during the month of June | (D) Earth ceases rotating on its axis for one week in June |

Q.4 During an El Niño Southern Oscillation (ENSO) event, what ocean-atmospheric anomaly occurs in the tropical Pacific, and how does it typically impact the Indian monsoon?

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| <p>(A) Abnormal warming of sea surface temperatures in the eastern Pacific shifts convective rainfall eastward, weakening the Walker circulation and often causing deficient rainfall or drought in India</p> | <p>(B) Cooling of the Pacific ocean doubles rainfall over the Thar desert</p> |
| <p>(C) El Niño causes permanent snow cover across all of southern India</p> | <p>(D) El Niño has zero impact on global atmospheric circulation</p> |

Q.5 Why does the leeward side of the Western Ghats (such as the Deccan Plateau / Marathwada / Vidarbha) receive significantly less rainfall than the windward coastal Konkan plains?

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| <p>(A) Moisture-laden winds ascend the windward slopes, cool adiabatically, and precipitate heavily; as they descend the leeward slopes, they warm adiabatically, lowering relative humidity and creating a rain-shadow zone</p> | <p>(B) The Western Ghats are made of absorbent sponge that soaks up all water</p> |
| <p>(C) Clouds are legally forbidden from crossing mountain ranges</p> | <p>(D) The Arabian sea only evaporates towards the west</p> |

Q.6 What meteorological role do 'Western Disturbances' originating over the Mediterranean Sea play in northern Indian winter agriculture?

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| <p>(A) They bring light to moderate winter rainfall and high-altitude snowfall, which is indispensable for the growth and grain-filling of the Rabi wheat crop</p> | <p>(B) They cause extreme tropical heatwaves that destroy wheat seedlings</p> |
| <p>(C) They bring torrential monsoon floods that submerge paddy fields in December</p> | <p>(D) They cause all northern Indian rivers to dry up instantly</p> |

Q.7 Why does Mawsynram in Meghalaya receive the highest average annual precipitation on Earth (>11,800 mm)?

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| <p>(A) The funnel-shaped alignment of the Khasi, Garo, and Jaintia hills traps incoming Bay of Bengal monsoon winds, forcing intense, continuous orographic ascent</p> | <p>(B) It is situated directly beneath an active volcanic steam plume</p> |
| <p>(C) Mawsynram is located at the South Pole where ice melts continuously</p> | <p>(D) It is surrounded on all sides by 8,000-meter glaciers</p> |

Q.8 How does the seasonal migration of the Inter-Tropical Convergence Zone (ITCZ) explain the reversal of winds over the Indian subcontinent between summer and winter?

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| <p>(A) In summer, intense solar heating pulls the ITCZ north over the Ganga plain (~20°–25°N), drawing southern hemisphere winds across the equator; in winter, the ITCZ retreats south over the Indian Ocean</p> | <p>(B) The ITCZ remains permanently locked over the North Pole</p> |
| <p>(C) The ITCZ is an underground river that flows north in summer</p> | <p>(D) Earth's magnetic poles swap positions every six months</p> |

Q.9 What is the physical mechanism behind the 'Greenhouse Effect' in Earth's atmosphere?

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| <p>(A) Atmospheric greenhouse gases (CO_2, H_2O, CH_4) are transparent to incoming shortwave solar radiation, but absorb and re-emit outgoing longwave infrared terrestrial radiation</p> | <p>(B) Gases in the air reflect all solar light back into outer space immediately</p> |
| <p>(C) The atmosphere acts as a solid glass wall that prevents oxygen from escaping</p> | <p>(D) Nuclear reactions inside clouds heat the surface directly</p> |

Q.10 Why is the destruction of stratospheric ozone by chlorofluorocarbons (CFCs) particularly hazardous to living organisms on Earth?

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| <p>(A) Increased penetration of high-energy Ultraviolet-B (UV-B) radiation damages DNA molecules, causing skin cancers, cataracts, and disrupting oceanic phytoplankton food webs</p> | <p>(B) It causes the Earth's atmosphere to run out of breathable nitrogen</p> |
| <p>(C) It stops all green plants from performing photosynthesis</p> | <p>(D) It causes ocean water to freeze into dry ice</p> |

Q.11 How does distance from the sea (continentality) influence the annual range of temperature in Delhi compared to Mumbai?

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| <p>(A) Delhi experiences an extreme continental climate with burning summers and chilling winters; Mumbai experiences an equable maritime climate moderated by the high specific heat capacity of the ocean</p> | <p>(B) Delhi is located inside a cave, while Mumbai is in outer space</p> |
| <p>(C) Mumbai experiences sub-zero snowfall while Delhi has zero temperature variation</p> | <p>(D) Both cities have identical daily and annual temperature curves</p> |

Q.12 What is a 'temperature inversion' in the troposphere, and what environmental hazard does it create in northern Indian cities during winter?

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| <p>(A) A condition where warm air caps cooler surface air, trapping particulate pollution and smoke beneath a stable layer to create dense toxic smog</p> | <p>(B) A situation where air temperature reaches absolute zero (-273 °C)</p> |
| <p>(C) A sudden heatwave that evaporates all groundwater</p> | <p>(D) A reversal where oxygen turns into liquid mercury</p> |

Q.13 Why does the retreat of the monsoon in October and November in northern India bring oppressive weather locally termed 'October Heat'?

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| <p>(A) Clear cloudless skies, high daytime temperatures, and high residual soil humidity create sultry, uncomfortable conditions</p> | <p>(B) Volcanic lava eruptions occur across the Gangetic plain in October</p> |
| <p>(C) The sun approaches within 1 million kilometers of the Earth in October</p> | <p>(D) All trees shed their leaves and absorb all atmospheric moisture</p> |

Q.14 How does the massive Tibetan Plateau act as an elevated thermal engine that drives the South Asian monsoon circulation?

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| <p>(A) Its immense altitude (~4,500 m) and intense summer solar heating create a powerful mid-tropospheric low pressure and rising air column that pulls maritime air from the Indian Ocean</p> | <p>(B) It acts as a giant refrigerator that cools the air to absolute zero</p> |
| <p>(C) It spins rapidly on a geological axle to blow wind across Asia</p> | <p>(D) It absorbs all carbon dioxide and converts it to pure methane</p> |

Q.15 Why are tropical cyclones originating over the Bay of Bengal during October and November particularly destructive to the deltaic coasts of Odisha, Andhra Pradesh, and West Bengal?

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| <p>(A) High sea surface temperatures (>26.5 °C) fuel intense cyclonic storms that generate catastrophic storm surges pushing sea water several kilometers inland across low-lying deltas</p> | <p>(B) The cyclones freeze all farmland into permafrost</p> |
| <p>(C) They drop molten sulfur rocks onto city streets</p> | <p>(D) They reverse the gravitational field along the coastline</p> |

Q.16 What is the fundamental difference between 'weather' and 'climate'?

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| (A) Weather describes day-to-day atmospheric conditions at a specific time and place; climate is the statistical average and pattern of weather over a long period (typically 30+ years) | (B) Weather applies to continents; climate applies to single villages |
| (C) Weather is measured with satellites; climate is measured with kitchen thermometers | (D) There is no difference; the terms are completely interchangeable |
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Q.17 Why does the Thar Desert in Rajasthan receive scanty rainfall despite lying in the path of the Arabian Sea branch of the southwest monsoon?

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| (A) The Aravalli mountain range runs parallel to the direction of incoming monsoon winds, offering no topographic barrier to force orographic ascent, and high heat prevents cloud condensation | (B) The Thar desert is surrounded by 10,000-meter high glass domes |
| (C) The Arabian sea branch of the monsoon carries zero moisture | (D) Winds are repelled by magnetic minerals in desert sand |
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Q.18 How do human activities like deforestation and coal-fired thermal power generation enhance the greenhouse effect?

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| (A) Deforestation destroys carbon sinks while fossil fuel combustion releases billions of tons of sequestered carbon dioxide into the active atmospheric carbon cycle | (B) Fossil fuels absorb atmospheric oxygen and turn it into solid ice |
| (C) Cutting trees increases stratospheric ozone levels beyond safe limits | (D) Coal combustion creates permanent gravitational anomalies |
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Q.19 What is the 'albedo' of an Earth surface, and why does melting Arctic sea ice create a dangerous positive climate feedback loop?

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| (A) Albedo is the fraction of solar radiation reflected by a surface; melting reflective white ice exposes dark ocean water, which absorbs more heat and accelerates further melting | (B) Albedo is the depth of ocean trenches measured in fathoms |
| (C) Albedo measures the strength of magnetic compass needles | (D) Melting ice causes the sun to shine 24 hours a day everywhere |
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Q.20 Which atmospheric layer exhibits the highest kinetic temperature (exceeding 1,500 °C), yet would feel freezing cold to a human hand due to extremely low molecular density?

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|-------------------------|------------------------|
| (A) Thermosphere | (B) Troposphere |
| (C) Stratosphere | (D) Ozonosphere |
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Q.21 Why are trade winds in the sub-tropics associated with the formation of the world's great hot deserts (Sahara, Arabian, Thar, Atacama) along the western coasts of continents?

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| (A) They blow as offshore dry winds by the time they reach western margins, and cold ocean currents offshore stabilize the air, suppressing convective rainfall | (B) The western margins are located closer to the moon |
| (C) Deserts are formed exclusively by falling volcanic ash | (D) Trade winds stop blowing when they reach the western coast |
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Q.22 What role does the 'Southern Oscillation' (SO) play in global climatology?

- (A) It is a seesaw pattern of atmospheric surface pressure between the tropical eastern Pacific Ocean (Tahiti) and the Indian Ocean (Darwin, Australia)
- (B) It is the physical wobbling of Earth's crust caused by earthquakes
- (C) It is the rotation of the moon around the earth
- (D) It is the seasonal migration of polar penguins

Q.23 How does carbon sequestration in soil and reforestation help mitigate global atmospheric warming?

- (A) Plants absorb carbon dioxide via photosynthesis and store it as durable biomass and soil organic carbon, removing it from the greenhouse blanket
- (B) Trees physically block the sun's rays from reaching the earth
- (C) Forests transform carbon dioxide into solid gold crystals
- (D) Soil organic matter reflects all incoming solar light into outer space

Q.24 Why does high-altitude mountain climbing require supplemental bottled oxygen above 8,000 meters ('Death Zone') even though the atmosphere is still 21% oxygen?

- (A) Atmospheric pressure drops drastically with altitude, resulting in a dangerously low partial pressure of oxygen (pO_2) that cannot adequately diffuse across alveolar lung membranes
- (B) The percentage of oxygen drops to 0% above 3,000 meters
- (C) The atmosphere above 8,000 meters is composed entirely of chlorine gas
- (D) Gravity reverses above 8,000 meters, pushing air out of the lungs

Q.25 What is the primary objective of the global 'Paris Climate Agreement' adopted under the UNFCCC?

- (A) To hold the increase in global average temperature to well below 2 °C above pre-industrial levels and pursue efforts to limit it to 1.5 °C
- (B) To ban all agriculture and farming across the world
- (C) To force all countries to shut down their electrical grids permanently
- (D) To build giant air-conditioning units across the Sahara desert

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET B)

01	(A) (B) (C) (D)	02	(A) (B) (C) (D)	03	(A) (B) (C) (D)	04	(A) (B) (C) (D)	05	(A) (B) (C) (D)
06	(A) (B) (C) (D)	07	(A) (B) (C) (D)	08	(A) (B) (C) (D)	09	(A) (B) (C) (D)	10	(A) (B) (C) (D)
11	(A) (B) (C) (D)	12	(A) (B) (C) (D)	13	(A) (B) (C) (D)	14	(A) (B) (C) (D)	15	(A) (B) (C) (D)
16	(A) (B) (C) (D)	17	(A) (B) (C) (D)	18	(A) (B) (C) (D)	19	(A) (B) (C) (D)	20	(A) (B) (C) (D)
21	(A) (B) (C) (D)	22	(A) (B) (C) (D)	23	(A) (B) (C) (D)	24	(A) (B) (C) (D)	25	(A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET B)

Q. No.	Correct	Curricular Reference & Analytical Rationale
Q.1	(A)	The troposphere is warmed by longwave infrared radiation reradiated from Earth's surface; in the stratosphere, ozone photolysis ($\text{O}_3 + h\nu$) releases heat, creating a temperature inversion.
Q.2	(A)	According to Ferrel's Law, Earth's eastward rotation deflects northern hemisphere winds to their right; winds flowing south towards the equator become Northeast Trade Winds.
Q.3	(A)	The rapid withdrawal of the westerly jet stream from northern India and the simultaneous onset of the tropical easterly jet stream trigger the dynamic 'burst' of the monsoon.
Q.4	(A)	Warm eastern Pacific water suppresses the normal western Pacific convective cell; this disrupts the Walker Circulation and descending air anomalies weaken the Indian monsoon trough.
Q.5	(A)	Orographic rainfall leaves leeward slopes dry: descending air warms by compression at the dry adiabatic lapse rate, raising its saturation deficit and suppressing rain (rain shadow).
Q.6	(A)	Temperate shallow cyclonic depressions steered by the westerly jet stream bring vital 20–50 mm of winter rain to Punjab, Haryana, and UP, ensuring high Rabi wheat yields.
Q.7	(A)	The unique topography of the Meghalaya plateau forms a horseshoe funnel open to the south, forcing moisture-saturated Bay of Bengal winds to ascend abruptly and dump torrential rains.
Q.8	(A)	Solar declination shifts the equatorial thermal trough 25°N in July (Monsoon Trough); cross-equatorial trade winds deflected right by Coriolis become the Southwest Monsoon.
Q.9	(A)	Shortwave insolation passes through atmospheric gases to warm the crust; the crust reradiates longwave infrared, which is absorbed by asymmetric dipolar molecules (CO_2 , CH_4).
Q.10	(A)	Ozone photochemically absorbs UV-B (280–315 nm). Excess UV-B induces pyrimidine dimers in cellular DNA, promoting mutations, skin carcinomas, and marine ecosystem collapses.
Q.11	(A)	Water's high specific heat capacity buffers coastal areas (maritime moderation); interior continental locations like Delhi lack maritime moderation and exhibit extreme temperature swings.
Q.12	(A)	Radiative winter cooling leaves cold, dense air near the ground beneath warm air; this thermal inversion suppresses vertical atmospheric mixing, trapping vehicular emissions as toxic smog.
Q.13	(A)	As the monsoon trough retreats, skies clear, raising insolation; high soil moisture and rising temperatures generate stifling atmospheric humidity known as 'October Heat'.
Q.14	(A)	The Tibetan Plateau functions as an elevated heat source, generating strong sensible heat fluxes in summer that create upper-tropospheric anticyclogenesis and power the monsoon engine.
Q.15	(A)	Post-monsoon tropical depressions intensify over warm waters; their ferocious cyclonic winds and shallow coastal bathymetry produce storm surges that inundate flat deltaic plains.
Q.16	(A)	Weather represents short-term atmospheric fluctuations (hours/days); climate represents long-term statistical meteorological distributions over standard 30-year epochs.
Q.17	(A)	The Aravalli hills align parallel to the southwest monsoon winds without obstructing them, and intense desert heat lowers relative humidity, preventing condensation.
Q.18	(A)	Combustion of fossil reserves oxidized millions of years of stored carbon, raising atmospheric CO_2 from pre-industrial ~280 ppm to over 420 ppm today.
Q.19	(A)	Ice-albedo feedback is a self-reinforcing loop: high-albedo sea ice (~0.6–0.9) reflects sunlight; open water (albedo ~0.06) absorbs 94% of insolation, heating the ocean.

Q. No.	Correct	Curricular Reference & Analytical Rationale
Q.20	(A)	In the thermosphere, individual gas atoms absorb high-energy solar X-rays (high kinetic temperature), but molecular density is so low that heat cannot transfer effectively.
Q.21	(A)	Subtropical high pressure drives descending dry air; easterly trade winds shed moisture on eastern margins and blow dry offshore in the west, reinforced by cold ocean upwelling.
Q.22	(A)	The Southern Oscillation Index (SOI) measures the pressure gradient between Tahiti and Darwin; negative values signify El Niño and disrupted rainfall across the Indo-Pacific.
Q.23	(A)	Biological carbon sequestration captures atmospheric CO_2 and locks it into woody biomass and stable humus, reducing net atmospheric radiative forcing.
Q.24	(A)	While the fractional composition remains 21% O_2 , barometric pressure drops from 1013 hPa at sea level to ~330 hPa at 8,848 m, reducing oxygen partial pressure by two-thirds.
Q.25	(A)	The 2015 Paris Agreement sets a legally binding international target to limit global mean surface warming to well below 2 °C, ideally 1.5 °C, via Nationally Determined Contributions.